

1 (2.11) verification

The admissible weight property

$$(G(t, \cdot) * \rho(\cdot))(x) = (2\pi t)^{-d/2} \int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) \rho(y) dy \leq K_T \rho(x) \quad \forall x \in \mathbb{R}^d, K_T \in \mathbb{N}, t \in [0, T]$$

tells us that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq K_T \quad t \in [0, T]. \quad (1.1) \quad \boxed{\text{aw}}$$

The expression for K_t should be

$$K_t := \sup_{x \in \mathbb{R}^d} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)}$$

The limit in (1.1) gives a sufficient (also necessary) condition for K_t to be finite, but in general, this limit is not equal to the value of K_t .

I meant for K_T to mean $C_\rho(T)$. I made some corrections above

We want to show that

$$\exp(-a|x|^2) \leq K'_a \rho(x) \quad (1.2) \quad \boxed{\text{con}}$$

Why does (1.1) imply (1.2)?

(1.4) and (1.1) imply (1.3) which implies (1.2)

and it suffices to show that

$$\lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq K'_a. \quad (1.3) \quad \boxed{\text{con1}}$$

since then

$$\exp(-a|x|^2) \leq (K'_a + \epsilon) \rho(x) \quad |x| > R_\epsilon$$

and by the positivity and continuity of both $\exp(-a|x|^2)$ and $\rho(x)$, we know that there exists some positive constant $K(a, \epsilon)$ such that

$$\exp(-a|x|^2) \leq K'(a, \epsilon) \rho(x) \quad x \leq R_\epsilon$$

and by then letting $K(a, \epsilon) = \max\{K'(a, \epsilon), (K'_a + \epsilon)\}$ then

$$\exp(-a|x|^2) \leq K(a, \epsilon) \rho(x) \quad x \in \mathbb{R}^d$$

which is what we want to show.

In order to show (1.3), it suffices to show that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\exp(-a|x|^2)} = \infty \quad (1.4) \quad \boxed{\text{con2}}$$

because this would mean that $\exp(-a|x|^2)$ is much smaller than $(G(t, \cdot) * \rho(\cdot))(x)$ for all large $|x|$ and since (1.1) is true then this would mean that (1.3) would be true as well by squeeze theorem,

$$0 \leq \lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq \lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq K_t$$

(1.4) is proved as follows: first recall that $|\rho(x)| < M$ for some $M \in \mathbb{N}$. Then,

$$0 \leq \frac{\int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) \rho(y) dy}{\exp(-a|x|^2)} \leq \frac{M \int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) dy}{\exp(-a|x|^2)}$$

$$\begin{aligned}
&= \frac{M * \sqrt{2\pi t}}{\exp(-a|x|^2)} \\
&\rightarrow \infty \quad \text{as } |x| \rightarrow \infty
\end{aligned}$$

Thus (1.4) is proved which means that (1.2) is also true.

Can you give an explicit form of $C_\rho(T)$ so that we can check whether $\sup_{T>0} C_\rho(T) < \infty$, or your K_t ? If not, can you derive some sufficient conditions that guarantee this property?

In the above wrote K_t to mean the same thing as $C_\rho(T)$. However in general there is no explicit form for this constant and also in general, $\sup_{T \geq 0} K_T \not\leq \infty$. For example, when we consider the weight $\rho(x) = \exp(-|x|)$ in $\mathbb{R}$ then we can show that $(\bar{G}(t, \cdot) * \exp(-|\cdot|))(x) \leq 2 \exp(t/2) \exp(-|x|)$ and clearly $\sup_{t \geq 0} 2 \exp(t/2) = \infty$.

This is a good example. We should not expect $\sup_{T>0} C_\rho(T) < \infty$. Actually, you may want to check eq. (4.4) of my AOP paper with Robert 2015 on p. 3040.

2 Mistake involving the $G(t, x) \leq KG(1, x)$ inequality

During a meeting we had, we proved that $G(t, x) \leq KG(1, x)$ where the constant K does not depend on x or t but we made a mistake in doing so. What we did is the following (for now we work in $\mathbb{R}$):

$$\begin{aligned}
\frac{G(t+1, x)}{G(1, x)} &= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2(t+1)} + \frac{|x|^2}{2}\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \left(-\frac{1}{t+1} + 1\right)\right) \quad \text{mistake was made here when factoring} \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \frac{t}{t+1}\right) \\
&\leq K
\end{aligned}$$

However if we fix the mistake, what we should get is that

$$\begin{aligned}
\frac{G(t+1, x)}{G(1, x)} &= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2(t+1)} + \frac{|x|^2}{2}\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(-\frac{|x|^2}{2} \left(\frac{1}{t+1} - 1\right)\right) \\
&= \sqrt{\frac{1}{t+1}} \exp\left(\frac{|x|^2}{2} \frac{t}{t+1}\right) \\
&\not\leq K
\end{aligned}$$

Let's remove the property $\sup_{t \geq 0} \mathbb{E} \left(|u(t, \cdot)|_\rho^2 \right) < \infty$ unless you can show that $C_\rho(T)$ is bounded in T .

3 Mistake that $G(t + t_0, x)$ is bounded above by $G(t, x)$ for $t > t_0$

$$\begin{aligned}
\frac{G(t + t_0, x)}{G(t_0, x)} &= \sqrt{\frac{t}{t + t_0}} \exp\left(-\frac{|x|^2}{2(t + t_0)}\right) \exp\left(\frac{|x|^2}{2t_0}\right) \\
&= \sqrt{\frac{t_0}{t + t_0}} \exp\left(\frac{|x|^2}{2(t + t_0)}\right)
\end{aligned}$$

and this is not bounded above by a constant independent of x and t .

This makes more sense!

4 Admissible Weight

4.1 $\rho(x) = \exp(-|x|)$

Let $\rho(x) = \exp(-|x|)$.

$$\begin{aligned}
 \sqrt{2\pi t} (G(t, \cdot) * \rho)(x) &= \int_{\mathbb{R}} \exp\left(-\frac{|y|^2}{2t}\right) \exp(-|x-y|) dy \\
 &\leq \int_{\mathbb{R}} \exp\left(-\frac{|y|^2}{2t}\right) \exp(-|x| + |y|) dy \quad \text{since } -|x-y| \leq -|x| + |y| \\
 &= \exp(-|x|) \int_{\mathbb{R}} \exp\left(-\frac{|y|^2}{2t}\right) \exp(|y|) dy \\
 &= \exp(-|x|) \int_{\mathbb{R}} \exp\left(-\frac{|y|^2 - 2t|y|}{2t}\right) dy \\
 &= \exp(-|x|) \exp(t/2) \int_{\mathbb{R}} \exp\left(-\frac{(|y| - t)^2}{2t}\right) dy \\
 &= \exp(-|x|) \exp(t/2) \sqrt{2\pi t} \left[\operatorname{erf}\left(\sqrt{\frac{t}{2}}\right) + 1 \right]
 \end{aligned}$$

and therefore

$$(G(t, \cdot) * \rho)(x) \leq \rho(x) \exp(t/2) \left[\operatorname{erf}\left(\sqrt{\frac{t}{2}}\right) + 1 \right].$$

Now fix $T > 0$. Since $\exp(t/2) \left[\operatorname{erf}\left(\sqrt{t/2}\right) + 1 \right]$ is increasing in t , then for every $t \in [0, T]$ we have that

$$(G(t, \cdot) * \rho)(x) \leq \exp(T/2) \left[\operatorname{erf}\left(\sqrt{\frac{T}{2}}\right) + 1 \right] \rho(x)$$

which shows that $\rho(x) = \exp(-|x|)$ is admissible with $C_\rho(T) = \exp(T/2) \left[\operatorname{erf}\left(\sqrt{\frac{T}{2}}\right) + 1 \right]$, which is the sharpest such constant for the particular ρ given in this example.

4.2 $\rho(x) = \exp(-|x|^2)$

In this case

$$\rho(x) = \sqrt{\pi} G(1/2, x)$$

and so we see that

$$(G(t, \cdot) * \rho)(x) = \sqrt{\pi} (G(t, \cdot) * G(1/2, \cdot))(x) = \sqrt{\pi} G(t + 1/2, x) = \frac{1}{\sqrt{2t+1}} \exp\left(\frac{-|x|^2}{2t+1}\right). \quad (4.1)$$

$\mathbf{E} : \exp(x^2)$

I claim that for any $T > 0$, that there does not exist a constant $K(T)$ such that

$$(G(t, \cdot) * \rho)(x) \leq K(T) \exp(-|x|^2) \quad \forall t \in [0, T] \text{ and } x \in \mathbb{R}.$$

BWOC, suppose there did. Then for $T = 4$, there exists some constant $K(4)$ such that for any $t \in [0, 4]$,

$$\frac{1}{\sqrt{2t+1}} \exp\left(\frac{-|x|^2}{2t+1}\right) \leq K(4) \exp(-|x|^2).$$

Note that for any $0 \leq t \leq 4$ and $x \in \mathbb{R}$,

$$\frac{1}{3} \exp\left(\frac{-|x|^2}{2t+1}\right) \leq \frac{1}{\sqrt{2t+1}} \exp\left(\frac{-|x|^2}{2t+1}\right)$$

and so we must have

$$\frac{1}{3} \exp\left(\frac{-|x|^2}{2t+1}\right) \leq K(4) \exp(-|x|^2).$$

This further implies that

$$\begin{aligned} 1 &\leq 3K(4) \exp\left(\frac{|x|^2}{2t+1} - |x|^2\right) \\ &= 3K(4) \exp\left(\frac{|x|^2 - 2t|x|^2 - |x|^2}{2t+1}\right) \\ &= 3K(4) \exp\left(\frac{-2t|x|^2}{2t+1}\right) \end{aligned}$$

and this must hold for all $t \in [0, 4]$ and $x \in \mathbb{R}$. However, as $|x| \rightarrow \infty$ then

$$3K(4) \exp\left(\frac{-2t|x|^2}{2t+1}\right) \rightarrow 0$$

and so

$$1 \leq 3K(4) \exp\left(\frac{-2t|x|^2}{2t+1}\right) \quad (4.2) \quad \boxed{\text{E:contradict.}}$$

cannot hold for all $x \in \mathbb{R}$ which is a contradiction. Thus $\exp(-|x|^2)$ is not admissible.

4.3 $\rho(x) = \exp(-|x|^b)$

Suppose that $b \in (1, 2)$ and that $\rho(x) = \exp(-|x|^b)$.

$$\begin{aligned} (G(t, \cdot) * \rho)(x) &= \int_{\mathbb{R}} G(t, x-y) \exp(-|y|^b) dy \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_{\mathbb{R}} G(t, x-y) |y|^{nb} dy \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \mathbb{E}(|N|^{nb}) \end{aligned}$$

where above N is a normal random variable with mean x and standard deviation t . Note that this can't technically be done because the power series for $\exp(x)$ does not converge uniformly on the whole real line, however doing this seems to work as can be seen in Section 4.4 below. According to [3], this moment is given by

$$\mathbb{E}(|N|^{nb}) = (2t)^{nb/2} \frac{\Gamma\left(\frac{nb+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{nb}{2}, \frac{1}{2}; -\frac{x^2}{2t}\right)$$

where ϕ is the Kummer Confluent Hypergeometric function

$${}_1F_1(\alpha, \gamma; z) = \phi(\alpha, \gamma; z) = \sum_{k=0}^{\infty} \frac{\alpha^{\bar{k}}}{\gamma^{\bar{k}}} \frac{z^k}{k!} \quad \text{and} \quad z^{\bar{k}} = \frac{\Gamma(z+k)}{\Gamma(z)}.$$

Plugging this in above gives us that

$$(G(t, \cdot) * \rho)(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (2t)^{nb/2} \frac{\Gamma\left(\frac{nb+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{nb}{2}, \frac{1}{2}; -\frac{x^2}{2t}\right) \quad (4.3) \quad \boxed{\text{E:exp}(x^{\wedge}b)}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (2t)^{nb/2} \frac{\Gamma\left(\frac{nb+1}{2}\right)}{\sqrt{\pi}} \sum_{k=0}^{\infty} \frac{(-nb/2)^{\bar{k}} \left(\frac{-x^2}{2t}\right)^k}{(1/2)^{\bar{k}} k!} \quad (4.4)$$

Note that it seems that (4.3) is only true when $b \in \mathbb{N}$ (see Section 4.5 and equation (4.6)).

If it works for $b \in \mathbb{N}$, it should work for fractions as well. I can show you some numerical evidence that why the case $b > 1$ won't work during our meeting. We may not need to spend too much time here unless we can really prove that ρ with $b > 1$ is not admissible.

4.4 Special case when $x = 0$ and $b = 2$

If we set $b = 2$ and $x = 0$ in (4.3) then we should be able to recover (4.1) with x set to 0, which is

$$\frac{1}{\sqrt{2t+1}}.$$

Letting $b = 2$ and $x = 2$ in (4.3) gives us

$$\sum_{n=0}^{\infty} \frac{(2n)!(-2t)^n}{4^n (n!)^2} \quad (4.5)$$

and plugging this into Wolfram gives us that

$$\sum_{n=0}^{\infty} \frac{(2n)!(-2t)^n}{4^n (n!)^2} = \frac{1}{\sqrt{2t+1}} \quad \text{whenever } |t| \leq 1/2.$$

Another thing to note is that the Borel Regularization of the sum (4.5) is also equal to $1/\sqrt{2t+1}$ for any $t > 0$. This can be easily checked by plugging

`Sum[(-2t)^k k(2k)!/(4^k (k!)^2), {k, 0, Infinity}, Regularization -> "Borel"]`

into Mathematica.

Its possible that we may be able to use this technique to calculate (4.3) for more general x and b . In fact, what I believe that we may be able to show is that if we fix any $t > 0$ then $(G(t, \cdot) * |\cdot|^b)(x)$ will decay much faster than $|x|^b$ as $|x| \rightarrow \infty$.

I can show you during our meeting this week that the case $b = 2$ won't work.

4.5 More Mathematica Calculations.

I am unsure that Borel Regularization can be used. For example, I considered $b = 3/2$, $t = 1/2$ and $x = 0$ and if we plug this into (4.3) then this gives us

$$(G(1/2, \cdot) * \rho)(0) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{\Gamma\left(\frac{n(3/2)+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{(3/2)n}{2}, \frac{1}{2}; 0\right)$$

however we get different answers when using Mathematica to calculate this and also this

$$\int_{\mathbb{R}} G(1/2, y) \exp\left(-|y|^{(3/2)}\right) dy.$$

The codes used were

`Sum[((-1)^n Gamma[((3/2)*n+1)/2] Hypergeometric1F1[(-(3/2)*n/2), 1/2, 0])/(Sqrt[Pi] n!),
{n, 0, Infinity}, Regularization -> "Borel"]`

S:SpecCase

E:specialcas

S:MMC

and

`Integrate[(1/Sqrt[Pi])Exp[-Abs[y]^2]Exp[-Abs[y]^(3/2)],{y,-Infinity,Infinity}]`.

However, it seems that the following is true for every $b \in \mathbb{N}$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{\Gamma\left(\frac{n(b)+1}{2}\right)}{\sqrt{\pi}} \phi\left(-\frac{(b)n}{2}, \frac{1}{2}; 0\right) = \int_{\mathbb{R}} G(1/2, y) \exp(-|y|^b) dy \quad (4.6)$$

E:BRsumrep

where the above sum is the Borel regularization.

Lastly, what I wished to be able to do what calculate

$$\int_{\mathbb{R}} G(t, x - y) \exp(-|y|^b) dy$$

on Mathematica for arbitrary t, x and b but I was not able to do so. The reason it gave me was that it took too much computational time for my account so maybe you could do it with your account. However to simplify this, we could just fix a t value, say $t = 1/2$ and this simplifies the integral to

$$\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^b) dy.$$

However I still cant calculate this on Mathematica. What I believe to be true is that if we assume by contradiction that for $T = 1/2$, there exists some constant $K(1/2)$ such that for any $0 \leq t \leq 1/2$ and $x \in \mathbb{R}$

$$\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^b) dy \leq K(1/2) \exp(-|x|^b)$$

then we will come up with a contradiction. If we use Mathematica, we see that for the case $t = 1/2$ and $b = 2$ then

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^2) dy}{\exp(-|x|^2)} = \infty$$

which again shows that $\exp(-|x|^2)$ is not admissible. But Mathematica would not calculate the corresponding limit for any particular $b \in (1, 2)$. For example, I tried taking $b = 3/2$, which is the limit

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \exp(-|x - y|^2) \exp(-|y|^{3/2}) dy}{\exp(-|x|^{3/2})}$$

and the codes I used were:

`f[x_] := Integrate[(1/Sqrt[Pi])Exp[-Abs[x-y]^2]Exp[-Abs[y]^(3/2)],{y,-Infinity,Infinity}]`

`Limit[f[x]Exp[Abs[x]^(3/2)],x ->Infinity]`

but Mathematica would time out each time I tried.

Here you might want to try “NIntegrate” instead of “Integrate”.

5 Extending the results to the fractional heat equation.

The following equation driven by space-time white noise is studied in [2]

$$\begin{cases} \left(\frac{\partial}{\partial t} - {}_x D_{\delta}^a\right) u(t, x) = \rho(u(t, x)) \dot{W}(t, x), & t > 0, \text{ and } x \in \mathbb{R} \\ u(0, \cdot) = \mu(\cdot). \end{cases}$$

The following moment bounds are proven there:

$$\|u(t, x)\|_p^2 \leq \begin{cases} J_0^2(t, x) + ([\bar{\varsigma} + J_0^2] \star \bar{\mathcal{K}})(t, x), & p = 2 \\ 2J_0^2(t, x) + ([\bar{\varsigma} + 2J_0^2] \star \hat{\mathcal{K}}_p)(t, x), & p > 2, \end{cases} \quad (5.1)$$

E:fracheatsp

and $\mathcal{K}$ has the following upper bound

$$\mathcal{K}(t, x) \leq \frac{C}{t^{1/a}} \delta G_a(t, x) \left(1 + t^{1/a} \exp(\gamma^{a^*} t)\right).$$

5.1 Discussion.

1. We run into the issue with this bound not being bounded in t
2. The space time convolution in the moment bound versus just a spacial convolution may cause problems. When dealing with the heat equation and a noise white in time and colored in space, the moment bound only has a spacial convolution.
3. We may be able to study equation (5.1) but with a noise that is white in time and colored in space and get a result similar to the heat equation case and use this to show the existence of an invariant measure.

6 $\Upsilon(\beta)|_{\beta=0} < \infty$ versus the spectral density integrability condition in [TZ98] (hypothesis 2.1).

Hypothesis 2.1 [TZ98] says that the spectral density $\hat{f}$ must satisfy the property that

$$\hat{f} \in L^p(\mathbb{R}^d) \quad \text{for some } p \in (d/(d-2), \infty] \quad (6.1)$$

E:hyp2.1

Equation (6.1) is not correct. Hypothesis 2.1 [TZ98] says that there exists a $p \geq 1$ such that $(1-1/p)(d/2) < 1$ which implies that $1 \leq p < d/(d-2)$. So equation (1.11) in the shortened version of the paper is not right. This also means that the two functions that I constructed down below are not correct as well.

Our assumption is that

$$\Upsilon(0) = (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(\xi)}{|\xi|^2} d\xi < \infty, \quad d \geq 3. \quad (6.2)$$

E:speccond

6.1 Assuming (6.2).

I believe that we may find a $\hat{f}$ that satisfies (6.2) but not (6.1). First, suppose (6.2) is true. Then we know that $\hat{f}(\xi)$ may have some relaxed decay properties at $|\xi| = \infty$ because of the help from the $|\xi|^{-2}$ term. In particular we can have decay as slow as $|\xi|^{-a}$ for $a \in (d-2, \infty)$ at infinity. At the same time, the $|\xi|^{-2}$ forces $\hat{f}$ to have growth no faster than $|\xi|^{-a}$ for $a \in (0, d-2)$ around the origin. So, if we assume $\hat{f}$ satisfies the condition (6.2) then $\hat{f}$ must be integrable around the origin but may not be integrable at infinity because $\hat{f}$ may have tails that decay as slow as $|\xi|^{-a}$ for $a \in (d-2, d)$.

Considering the above, it seems as if we assume $\hat{f}$ satisfies (6.2), then $\hat{f}$ may not be able to satisfy the condition (6.1). First, let $d \geq 3$ and consider $0 < \delta < K$ where K will be determined below. Now consider $\hat{f}$ such that $\hat{f}$ behaves like $|\xi|^{-(d-2-\delta)}$ around the origin and $|\xi|^{-(d-2+\delta)}$ at infinity. BWOC suppose that $\hat{f}$ satisfies condition (6.1). Then to take care of the thick tails, we would need p to be such that

$$p > \frac{d}{d-2+\delta} \quad (6.3)$$

but recall that $p > d/(d-2)$ so we must choose a $\gamma > 0$ such that

$$\frac{d+\gamma}{d-2+\delta} > \frac{d}{d-2}$$

and thus we need

$$p \geq \frac{d+\gamma}{d-2+\delta}. \quad (6.4)$$

E:pcon

However, at the same time we still need to maintain integrability around the origin. But we will now see that if we choose $\delta > 0$ small enough that we will lose integrability around the origin. Note that because of (6.4) we may say that

$$p(d-2-\delta) \geq \frac{(d+\gamma)(d-2-\delta)}{d-2+\delta}$$

and if we set

$$\frac{(d+\gamma)(d-2-\delta)}{d-2+\delta} \geq d$$

then this is satisfied when

$$\frac{d-2-\delta}{d-2+\delta} \geq \frac{d}{d+\gamma} \quad (6.5)$$

E:hyp2.1fail

and since $\delta > 0$ is arbitrary and

$$\lim_{\delta \rightarrow 0} \frac{d-2-\delta}{d-2+\delta} = 1 > \frac{d}{d+\gamma}$$

then there exists a $K > 0$, independent of p , such that for any $\delta \in (0, K)$ that (6.5) is true. And therefore this implies that

$$p(d-2-\delta) \geq d$$

and so $\hat{f}$ cannot be integrable around the origin which means that $\hat{f}$ satisfies (6.2) but not (6.1).

6.2 Assuming (6.1).

I believe that an additional property of the spectral measure, $\hat{f}$, is that $\hat{f}(\xi) < \infty$ if and only if $t \neq 0$. This is assumed in [1].

So let's assume that $\hat{f}$ blows up at the origin. Also assume that $\hat{f}$ satisfies the condition (6.1). I believe that even if we assume that $\hat{f}$ must blow up at the origin, then we may still construct an example that satisfies (6.1) but not (6.2) which would conclude that not one condition is more general than the other.

Let's consider an $\hat{f}(\xi)$ such that around the origin, $\hat{f}$ behaves like $|\xi|^{-(d-2-\delta)}$ for some arbitrary $\delta > 0$. Suppose the tail of $\hat{f}$ behaves like $|\xi|^{-(d-2)}$. We want to find a $p > d/(d-2)$ such that $\hat{f} \in L^2(\mathbb{R}^d)$. Because of the behavior around the origin, we would need that

$$p < \frac{d}{d-2-\delta}.$$

But

$$\frac{d}{d-2-\delta} > \frac{d}{d-2}$$

and so we must restrict our choice of p to be such that

$$\frac{d}{d-2} < p < \frac{d}{d-2-\delta}. \quad (6.6)$$

E:pinterval

Notice that because of the tail behavior, that we only need

$$p(d-2) > d$$

which means that any p satisfying (6.6) will make the tail $L^p(\mathbb{R}^d)$ integrable. Therefore, for such an $\hat{f}$ with origin and tail behavior as described above, we may be able to find a p such that $\hat{f} \in L^p(\mathbb{R}^d)$.

But this $\hat{f}$ will not satisfy the condition (6.2) because the tail behavior of $|\xi|^{-2}\hat{f}(\xi)$ will be $|\xi|^{-d}$, and this is not integrable.

Your construction/proof is convincing! This second case suggests that [TZ98] has to contain some flaw! Can you apply my results in [CK19] to show that the case with

$$\hat{f}(\xi) = \frac{1}{1 + |\xi|^{d-2}}, \quad d \geq 3$$

is a legal spectral density (namely, nonnegative and nonnegative definite), and solution is intermittent (namely, no phase transition).

$$7 \quad \tilde{\Gamma} = |\widehat{\gamma^{1/2}}| * |\widehat{\gamma^{1/2}}|$$

Simplifying the function

$$\tilde{\Gamma} := |\widehat{\gamma^{1/2}}| * |\widehat{\gamma^{1/2}}| = \mathcal{F}(|\gamma^{1/2}|) * \mathcal{F}(|\gamma^{1/2}|) \quad (7.1) \quad \boxed{\text{E:GammaTilde}}$$

gives us up to a constant

$$\tilde{\Gamma}(\xi) = \mathcal{F}(|\gamma|)(\xi)$$

Last time we met you mentioned to look at

$$g(x) = \prod_{i=1}^3 1_{[-1,1]}(x_i), \quad \mathcal{F}g(\xi) = 8 \prod_{i=1}^3 \frac{\sin(\xi_i)}{\xi_i}.$$

Is g supposed to represent γ ? If so then I think this is a problem. Let f be the spatial correlation function. Since γ is the spectral density, then up to a constant, we would have

$$f(x) = \mathcal{F}(g)(x) = 8 \prod_{i=1}^3 \frac{\sin(x_i)}{x_i}.$$

but this is a problem because the correlation function needs to be nonnegative and nonnegative definite and $\sin(x)/x$ is not nonnegative.

I think we made another mistake as well. We defined $q(x) = (g * g)(x)$ and noted that $\mathcal{F}q = (\mathcal{F}g)^2$. However, we are never taking the Fourier transform of $g * g$. (7.1) is a convolution of Fourier transforms, not a Fourier transform of convolutions. Using (7.1), we wont get the

$$\left| \frac{\sin(x)}{x} \right|$$

term since by using (7.1),

$$\tilde{\Gamma}(x) = \mathcal{F}(|g(\cdot)|)(x) = \mathcal{F}(g(\cdot))(x) = \prod_{i=1}^3 \frac{\sin(x_i)}{x_i}.$$

In our meeting we did

$$\sqrt{\mathcal{F}q} = \sqrt{(\mathcal{F}g)^2} = 8 \prod_{i=1}^3 \left| \frac{\sin(x_i)}{x_i} \right|$$

but I don't think this should be done.

Let $f(x) = (g * g)(x)$. h is a tent function as I drew in the paper during our meeting. Please show that h is nonnegative and nonnegative definite. Moreover, how would you find $\sqrt{\mathcal{F}f(\xi)}$?

Isn't the condition

$$\tilde{\Gamma} = \mathcal{F}(|\gamma^{1/2}|) * \mathcal{F}(|\gamma^{1/2}|), \quad \frac{\Gamma(d/2 - 1)}{4\pi^{d/2}} \int_{\mathbb{R}^d} \tilde{\Gamma}(\xi) |\xi|^{d-2} d\xi < \infty \quad (7.2) \quad \boxed{\text{E:GamTilCon}}$$

the same thing as Dalang's condition? I don't think the absolute value adds any restriction because up to a constant we have:

$$\begin{aligned} \tilde{\Gamma} &= \mathcal{F}(|\gamma^{1/2}|) * \mathcal{F}(|\gamma^{1/2}|) \\ &= \mathcal{F}\left(|\gamma^{1/2}| \cdot |\gamma^{1/2}|\right) \\ &= \mathcal{F}(|\gamma|) \\ &= \mathcal{F}(\gamma) \\ &= f \end{aligned}$$

where above f is the correlation function and γ is the spectral density and hence both are nonnegative and nonnegative definite. Therefore, (7.2) can be written as the following:

$$\int_{\mathbb{R}^d} \tilde{\Gamma}(\xi) |\xi|^{d-2} d\xi = \int_{\mathbb{R}^d} f(\xi) |\xi|^{d-2} d\xi$$

which is the same thing as Dalang's condition.

8 Meaning of $\mathcal{F}(\gamma^{1/2})$.

We discussed last meaning that equation (3.4) [TZ98], which says

$$\tilde{\Gamma} := |\widehat{\gamma^{1/2}}| * |\widehat{\gamma^{1/2}}|$$

means that we are calculating $\mathcal{F}(\gamma^{1/2})$ and then taking the absolute value and then doing the convolution. So in other words

$$\tilde{\Gamma} := |\mathcal{F}(\gamma^{1/2})| * |\mathcal{F}(\gamma^{1/2})|. \quad (8.1) \quad \boxed{\text{E:GamTil}}$$

We saw that this was the case by observing the proof of Theorem 3.3 [TZ98].

We then wanted to consider the case where the correlation function, f , is given by

$$f(x) = (2 - |x|)1_{[-2,2]}(x)$$

which means that

$$\hat{f}(\xi) = \gamma(\xi) = \frac{\sin^2(x)}{x^2}.$$

Because $\gamma(\xi)$ touches 0 an infinite number of times then we have an infinite number of options to choose from for $\sqrt{\gamma(\xi)}$. Notice that (8.1) does not tell us which branch of $\sqrt{\gamma}$ to choose since the absolute value is not taken until after taking the Fourier transform. We considered the following two branches for $\sqrt{\gamma}$:

$$\frac{\sin(\xi)}{\xi} \quad \text{or} \quad \frac{|\sin(\xi)|}{|\xi|} \quad (8.2) \quad \boxed{\text{E:SqrtGamma}}$$

however it seems like we could also choose

$$\frac{|\sin(\xi)|}{|\xi|} K(\xi) \quad (8.3) \quad \boxed{\text{E:SqrtGammaK}}$$

where

$$K(\xi) = \begin{cases} -1 & \xi \in (2\pi, 3\pi) \\ 1 & \text{otherwise} \end{cases}$$

and we could come up with an infinite number of choices for $\sqrt{\gamma}$ of the form (8.3). We wanted to choose the second choice from (8.2) but what tells us that this is the choice we are supposed to choose? It seems to me now that there are an infinite number of possibilities to define $\tilde{\Gamma}$ and note that if we just consider the two choices from (8.2) then we can get that

$$\tilde{\Gamma} = \left| \mathcal{F} \left(\frac{\sin(\cdot)}{\cdot} \right) \right| * \left| \mathcal{F} \left(\frac{\sin(\cdot)}{\cdot} \right) \right| \quad \text{or} \quad \tilde{\Gamma} = \left| \mathcal{F} \left(\left| \frac{\sin(\cdot)}{\cdot} \right| \right) \right| * \left| \mathcal{F} \left(\left| \frac{\sin(\cdot)}{\cdot} \right| \right) \right|$$

and I do not believe that these two choices for $\tilde{\Gamma}$ are equal.

This makes me believe that there is something not being said in this paper that tells us which branch of $\sqrt{\gamma}$ needs to be chosen. I think this has something to do with the centered equation right below equation (2.3) [TZ98] which says that

$$M(\varphi)\psi = \varphi(\mathcal{F}(\gamma^{1/2}) * \psi) \quad (8.4)$$

E:SqrtGammaM

where

$$X^x(t) = S(t)x + \int_0^t S(t-s)M(B(X^x(s)))dW_s$$

and

$$B(\phi)(\xi) = b(\xi, \phi(\xi)).$$

I will look through [DZ92a] and [DZ92b] to see if this has to do with the construction of the DaPrato integral.

9 Proof of Theorem 3.1 [TZ98]

The proof starts by saying that the result can be proven once we show that for any $\epsilon \in (0, 1)$ and $R > 0$ that there exists a compact set $\mathcal{K}(\theta) \subset L_\rho^2$ such that

$$\mathbb{P} \{X^{\hat{x}}(t) \in \mathcal{K}(\theta)\} \geq (1 - \epsilon) \mathbb{P} \{|X^{\hat{x}}(t-1)|_{\hat{\rho}} < R\} \quad \text{for all } t > 1. \quad (9.1)$$

E:3.1ineq

In **step 3** of this proof they define the set

$$\mathcal{K}(\theta) := \left\{ S(1)y + \frac{\sin(\alpha\pi)}{\pi} F_\alpha h : |y|_{\hat{\rho}} \leq \theta \text{ and } |h|_{L^q(0,1,L_\rho^2)} \leq \theta \right\} \quad (9.2)$$

E:KTheta

and they also mention that by the factorization theorem from [DZ96] that

$$X^x(1) = S(1)x + \frac{\sin(\alpha\pi)}{\pi} F_\alpha(Y^x(\cdot)). \quad (9.3)$$

E:Factorizat

They claim that since

$$\mathbb{P} \{X^x(1) \in \mathcal{K}(\theta)\} \geq (1 - k_3\theta^{-q}(1 + R^q)) \quad (9.4)$$

E:KTheta1

then we have

$$\mathbb{P} \{X^x(t) \in \mathcal{K}(\theta)\} \geq (1 - k_3\theta^{-q}(1 + R^q)) \mathbb{P} \{|X^x(t-1)|_{\hat{\rho}} < R\} \quad \text{for all } t > 1. \quad (9.5)$$

E:KThetaRecu

I am not sure why this true. I was thinking that it follows from something like the following:

$$\begin{aligned} \mathbb{P} \{X^x(t) \in \mathcal{K}(\theta)\} &= \mathbb{P} \left\{ X^x(t) \in \mathcal{K}(\theta) \mid |X^x(t-1)|_{\hat{\rho}} < R \right\} \mathbb{P} \{|X^x(t-1)|_{\hat{\rho}} < R\} \\ &\quad + \mathbb{P} \left\{ X^x(t) \in \mathcal{K}(\theta) \mid |X^x(t-1)|_{\hat{\rho}} \geq R \right\} \mathbb{P} \{|X^x(t-1)|_{\hat{\rho}} \geq R\}. \end{aligned}$$

However, we would want something like

$$R > |X^x(t-1)|_{\hat{\rho}} \quad (9.6)$$

E:RU**p**Bd

which implies that

$$\mathbb{P} \left\{ |X^x(t-1)|_{\hat{\rho}} \geq R \right\} = 0$$

and so

$$\mathbb{P} \{X^x(t) \in \mathcal{K}(\theta)\} = \mathbb{P} \left\{ X^x(t) \in \mathcal{K}(\theta) \mid |X^x(t-1)|_{\hat{\rho}} < R \right\} \mathbb{P} \left\{ |X^x(t-1)|_{\hat{\rho}} < R \right\}. \quad (9.7)$$

E:K**Theta**Cond

This can't be the case though because this would imply that the following function is bounded, which we do not know to be true:

$$(t, \omega) \mapsto |X^x(t-1)|_{\hat{\rho}}(\omega) \quad \text{for all } (t, \omega) \in (1, \infty) \times \Omega.$$

However, we do know that

$$\sup_{t > \alpha} \mathbb{E} [|X^x(t, \cdot)|_{\hat{\rho}}^2] < \infty$$

so maybe this helps? However the statement in (9.1) says that it needs to be for arbitrary $R > 0$ and (9.6) implies that R must be taken sufficiently big and thus R is not arbitrary.

Now just assuming the above is it true, can we say that the SHE is time homogeneous in the sense that

$$\mathbb{P} \left\{ X^x(t) \in \mathcal{K}(\theta) \mid |X^x(t-1)|_{\hat{\rho}} < R \right\} = \mathbb{P} \left\{ X^x(1) \in \mathcal{K}(\theta) \mid |X^x(0)|_{\hat{\rho}} < R \right\}?$$

Because if this is true then because of (9.4) we have that

$$\begin{aligned} \mathbb{P} \left\{ X^x(t) \in \mathcal{K}(\theta) \mid |X^x(t-1)|_{\hat{\rho}} < R \right\} &= \mathbb{P} \left\{ X^x(1) \in \mathcal{K}(\theta) \mid |X^x(0)|_{\hat{\rho}} < R \right\} \\ &\geq (1 - k_3 \theta^{-q} (1 + R^q)) \end{aligned}$$

and combining this with (9.7) gives us (9.5).

10 Two possible assumptions for the initial condition.

We have been deciding between which of the following two assumptions we want to assume:

1.

$$\sup_{t > 0} \int_{\mathbb{R}^d} (G(t, \cdot) * |\mu|)^2(x) \rho(x) dx < \infty \quad (10.1)$$

E:A**1**

2.

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |\mu|)(x) < K(t) \quad \text{for some } K(t) \in \mathbb{N}. \quad (10.2)$$

E:A**2**

You originally suggested Condition (10.1) because it gave a nice relation between the initial condition and the weight and that it may have been able to replace the admissible weight condition. However we saw last meeting that we still need to assume the admissible weight condition because of Proposition 2.1 [TZ98].

The other issue with Condition (10.1) is that

$$\sup_{t > 0} \int_{\mathbb{R}^d} (G(t, \cdot) * \delta_0)^2(x) \rho(x) dx = \infty.$$

In order to fix this issue with δ_0 , we discussed that we may potentially be able to replace Condition (10.1) with the following:

$$\sup_{t > \lambda} \int_{\mathbb{R}^d} (G(t, \cdot) * |\mu|)^2(x) \rho(x) dx < \infty \quad \text{for any } \lambda > 0. \quad (10.3)$$

E:A**1.1**

Now if we assume Condition (10.3), then for $\varphi \in L^2_\rho$, we will only be able to show that

$$\sup_{t>\lambda} \mathbb{E}(|u(t, \cdot)|_\rho) < \infty.$$

Using this, we will then only be able to show the following bounded in probability statement: for all $\epsilon > 0$ there exists an $R > 0$ such that

$$\mathbb{P}(|u(t, \cdot)|_\rho \geq R) < \epsilon \quad \text{for all } t \geq \lambda.$$

I have not verified this, but I believe that if we assume (10.3) then our version of (9.1) becomes: for $0 < t_0 \leq 1$ and $\epsilon \in (0, 1)$ and $R > 0$, there exists a compact set $\mathcal{K}(\theta) \subset L^2_\rho(\mathbb{R}^d)$ such that

$$\mathbb{P}\{X^{\hat{x}}(t) \in \mathcal{K}(\theta)\} \geq (1 - \epsilon)\mathbb{P}\{|X^{\hat{x}}(t - t_0)|_{\hat{\rho}} < R\} \quad \text{for all } t > \lambda + t_0 \quad (10.4)$$

E:Conjecture

If we assume all of this then we can prove that the laws

$$\{\mathcal{L}(u(t, \cdot))\}_{t>t_0+\lambda}$$

are tight and hence there exists an invariant measure. So we get the same conclusion as long as the conjecture (10.4) is correct.

11 I do not think (10.3) will work as an assumption.

Recall that we defined for $f(s, y) \in L^q(0, 1, L^2_\rho)$

$$(F^{t_0}_\alpha f)(x) := \int_0^{t_0} ds \int_{\mathbb{R}^d} dy (t_0 - s)^{\alpha-1} G(t_0 - s, x - y) f(s, y).$$

and

$$Y^\varphi(s, y) = \int_0^s \int_{\mathbb{R}^d} (s - r)^{-\alpha} G(s - r, y - z) b(u^\varphi(r, z)) W(dr, dz).$$

In Lemma 4.1, we proved that (there is a typo in our paper, the integral needs to go from 0 to t_0)

$$\mathbb{E} \int_0^{t_0} Y^\varphi(s, \cdot)|_\rho^q \leq |\varphi|_\rho^q. \quad (11.1)$$

E:YNorm

In order to prove this result we either need (10.1) or (10.2). If we are only going to assume (10.3) then all we will be able to say is that

$$\mathbb{E} \int_\lambda^{t_0} Y^\varphi(s, \cdot)|_\rho^q \leq |\varphi|_\rho^q \quad (11.2)$$

E:YNorm1.1

and this will not be enough in order to prove the conjecture (10.4).

This is because by Lemma 4.3 in our paper we have that

$$\begin{aligned} \frac{\sin(\alpha\pi)}{\pi} \int_0^t (t - s)^{\alpha-1} [G(t - s, \cdot) * Y(s, \cdot)](x) ds &= \frac{\sin(\alpha\pi)}{\pi} (F^{t_0}_\alpha Y^\varphi)(x) \\ &= \int_0^t \int_{\mathbb{R}^d} G(t - r, x - z) b(u(r, z)) W(dr, dz) \end{aligned}$$

and thus

$$u^\varphi(t_0, x) = (G(t_0, \cdot) * \varphi)(x) + \frac{\sin(\alpha\pi)}{\pi} (F^{t_0}_\alpha Y^\varphi)(x)$$

and if you examine the proof of Lemma 3.1, this factorization is very important and in particular, the inequality (11.1). I do not see how (11.2) could be used to replace it.

One last thing to add is that this problem does not present itself with the original assumption (10.2) so we may want to stick with that in order to apply the δ_0 initial condition. However, this means we can only used bounded initial conditions.

12 Tightness of $U(t_0, \cdot)$

Define

$$U(T, \cdot)(A) = \frac{1}{T} \int_{t_0}^{T+t_0} \mathcal{L}(u_{u_0}(t, \cdot))(A) dt \quad (12.1) \quad \boxed{\text{E:U}}$$

where u_0 is the initial condition and $A \in \mathcal{B}(L^2_\rho(\mathbb{R}^d))$. We see that $\mathcal{L}(u_{u_0}(t, \cdot))(A)$ defines a probability measure on $\mathcal{B}(L^2_\rho(\mathbb{R}^d))$. Because $u_{u_0}(t, x)$ is jointly continuous on $(0, \infty) \times \mathbb{R}^d$ then I believe that $t \mapsto \mathcal{L}(u_{u_0}(t, \cdot))(A)$ is continuous.

First I will show that (12.1) defines a probability measure. Lets consider a general measure space $(E, \mathcal{E})$ and let $\mathcal{M}_1(E)$ denote the set of probability measures on $(E, \mathcal{E})$. Define

$$P_T(A) := \frac{1}{T-a} \int_a^T p_t(A) dt, \quad (12.2) \quad \boxed{\text{E:IntProb}}$$

where $T > a \geq 0$, $t \in (a, T)$, $p_t \in \mathcal{M}_1(E)$, and $t \mapsto p_t(A)$ is continuous for each $A \in \mathcal{E}$. **Eq. (12.2) cannot be true. In other words:**

$$\begin{aligned} \mathbb{P}(U(T, \cdot) \in A) &= \mathbb{P}\left(\frac{1}{T} \int_{t_0}^{t_0+T} u(s, \cdot) ds \in A\right) \\ &\neq \frac{1}{T} \int_{t_0}^{t_0+T} \mathbb{P}(u(s, \cdot) \in A) ds. \end{aligned}$$

$$\begin{aligned} \mathbb{P}(U(T, \cdot) \in A) &:= U(T, \cdot)(A) \\ &= \frac{1}{T} \int_{t_0}^{T+t_0} \mathcal{L}(u_{u_0}(t, \cdot))(A) dt. \end{aligned}$$

So in (12.2), we let $p_t(A) = \mathcal{L}(u_{u_0}(t, \cdot))(A)$. However I am starting to think that all of this is unnecessary. Since the family of measures $\{\mathcal{L}(u_{u_0}(t, \cdot))(\circ)\}_{t>0}$ is tight then given any $\epsilon > 0$ we can find a compact set $K \in \mathcal{B}(L^2_\rho(\mathbb{R}^d))$ such that

$$\mathcal{L}(u_{u_0}(t, \cdot))(K^c) = \mathbb{P}[\omega \in \Omega : u_{u_0}(t, \cdot)(\omega) \in K^c] \leq \epsilon.$$

Then we immediately see that

$$\begin{aligned} U(T, \cdot)(K^c) &= \frac{1}{T} \int_{t_0}^{T+t_0} \mathcal{L}(u_{u_0}(t, \cdot))(K^c) dt \\ &\leq \frac{1}{T} \int_{t_0}^{T+t_0} \epsilon dt \\ &= \epsilon, \end{aligned}$$

So the family of probability measures $\{U(T, \cdot)(\circ)\}_{T>0}$ is tight. So above in (12.2), we let $p_t(A) = \mathcal{L}(u_{u_0}(t, \cdot))(A)$.

(12.2) defines a Probability Measure: First, consider the probability measures $p_1, \dots, p_n \in \mathcal{M}_1(E)$. It is easy to verify that $(p_1 + \dots + p_n)/n \in \mathcal{M}_1(E)$. Now suppose that for $T > a \geq 0$ and $t \in (a, T)$ that $p_t \in \mathcal{M}_1(E)$ and that $t \mapsto p_t(A)$ is continuous for each $A \in \mathcal{E}$. Then the Lebesgue integral in (12.2) exists for each $A \in \mathcal{E}$. It remains to show that $P_T \in \mathcal{M}_1(E)$. Note that

$$P_T(A) = \lim_{t_n \in P_n} \frac{1}{T-a} \sum_{t_k \in P_n} p_{t_k}(A) \Delta_n$$

where $\Delta_n = (T - a)/n$ and P_n is a partition of $[a, b]$ constructed from such Δ_n . From this we easily deduce that

$$P_T(E) = \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{t_k \in P_n} p_{t_k}(E) \Delta_n = \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{t_k \in P_n} \Delta_n = 1.$$

In addition, it is clear that $P_T(A) \geq 0$ so it remains to show countable additivity. Suppose $\{A_j\}_{j \geq 1} \in \mathcal{E}$ are disjoint. Then,

$$\begin{aligned} P_T \left(\sum_{j \geq 1} A_j \right) &= \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{t_k \in P_n} p_{t_k} \left(\sum_{j \geq 1} A_j \right) \Delta_n \\ &= \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{j \geq 1} \sum_{t_k \in P_n} p_{t_k}(A_j) \Delta_n \end{aligned}$$

where the last line follows because each $p_{t_k} \in \mathcal{M}_1(E)$. Since

$$\sum_{t_k \in P_n} p_{t_k}(A_j) \Delta_n \leq \sum_{t_k \in P_n} \Delta_n = 1,$$

then the dominated convergence theorem implies that

$$\begin{aligned} \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{j \geq 1} \sum_{t_k \in P_n} p_{t_k}(A_j) \Delta_n &= \sum_{j \geq 1} \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{t_k \in P_n} p_{t_k}(A_j) \Delta_n \\ &= \sum_{j \geq 1} p(A_j), \end{aligned}$$

and so P_T is countably additive which means $P_T \in \mathcal{M}_1(E)$.

Remark 12.1. It seems that the assumption $t \mapsto p_t(A)$ defines a continuous mapping may be unnecessary. It seems all we would really need is the existence of the Lebesgue integral in (12.2) which only requires $t \mapsto p_t(A)$ be measurable and bounded almost everywhere. DaPrato [DZ96] requires that $P_t(x, A) = u_x(t, \cdot)(A)$ be a stochastically continuous Feller semi-group in order for (12.1) to be a probability measure, which happens to be the case in our situation.

(12.2) is tight if the p_t are tight: Now let's further assume that the family of probability measures $\{p_t\}_{t \in [a, T]}$ is tight. Let $\epsilon > 0$ be given. Then there exists a compact set $K \in \mathcal{E}$ such that for each $t \in [a, T]$, $p_t(K^c) < \epsilon$ where $K^c = E \setminus K$. Then,

$$\begin{aligned} P_T(K) &= \frac{1}{T - a} \int_a^b p_t(K) dt \\ &= \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{t_k \in P_n} p_{t_k}(K) \Delta_n \\ &\leq \epsilon \lim_{t_n \in P_n} \frac{1}{T - a} \sum_{t_k \in P_n} \Delta_n \\ &= \epsilon, \end{aligned}$$

which implies that the family $\{P_T\}_{T \geq a}$ defined in (12.2) is tight.

13 Assumptions for Fubini's

If you examine the complete proof that I have written in the long version of the invariant measure paper, you will see that in order to apply Fubini's theorem, we will need to assume the following:

1. $\|G(t, x - \cdot)\varphi(\cdot)\|_{L^1(\mathbb{R}^d)} < \infty$,
2. $\mathcal{H}(s) = \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) < \infty$,
3. $\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)|\xi|^2\right) < \infty$,
4. $\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{2s}|\xi|^2\right) < \infty$.

The first item is true since we assume out initial measure, φ satisfies

$$\int_{\mathbb{R}^d} \exp(-a|x|^2)\varphi(x)dx < \infty.$$

The second item is also true by assumption. Lastly, the finiteness of $\mathcal{H}(s)$ implies that items 3 and 4 are true, which is proven in Lemma 4.2 of the longer version of the paper on GitHub.

14 Bessel kernal

Unfortunately, we may not consider the Riesz kernel in our setting due to the assumption that $\Upsilon(0) < \infty$. Indeed, for $0 < s < d$, the Riesz kernel is defined to be $f(x) = |x|^{s-d}$ and in order for

$$\Upsilon(0) = \int_{\mathbb{R}^d} \frac{f(x)}{|x|^{d-2}} dx = \int_{\mathbb{R}^d} \frac{1}{|x|^{2d-s-2}} < \infty,$$

one would need both $s-2 > d$ and $s-2 < d$, which clearly is not possible. However, we may consider the Bessel kernel, $G_s(x)$, which satisfies

$$(I - \Delta)^{-s/2} f = f * G_s,$$

where,

$$\mathcal{F}G_s(\xi) = \frac{1}{(1 + |\xi|^2)^{s/2}}.$$

Moreover, it can be shown (see [Gra14, Proposition 1.2.5]) that

1. $G_s(x) = \frac{(\pi)^{-d/2}}{\Gamma(s/2)} \int_0^\infty e^{-t} e^{-|x|^2/(4t)} t^{(s-d)/2} \frac{dt}{t}$,
2. $G_s(x) \leq c(s, d) (|x|^{s-d} + 1 + O(|x|^{s-d+2}))$, for $0 < s < d$ and $|x| \leq 2$,
3. $G_s(x) \leq C(s, d) e^{-|x|/2}$, when $|x| \geq 2$.

Hence, $G_s(x)$ exhibits exponential decay as $x \rightarrow \infty$, and so if we consider $d \geq 3$, $d-2 < s < d$, and $\alpha < 1/2$ then it is easy to see that both $\mathcal{H}(t) < \infty$ for any $t > 0$ and $\Upsilon(0) < \infty$.

Details for item 1 above. Consider the following definitions for the inverse Fourier transform:

$$\mathcal{F}_0^{-1} f(x) = \int_{\mathbb{R}^d} f(\xi) \exp(2\pi i \xi x) dx \quad \text{and} \quad \mathcal{F}^{-1} f(x) = \int_{\mathbb{R}^d} f(\xi) \exp(i \xi x) dx.$$

Hence,

$$\mathcal{F}_0^{-1} f\left(\frac{x}{2\pi}\right) = \mathcal{F}^{-1} f(x).$$

Before we continue we recall the following:

$$\text{if } f(\xi) = \exp(-\pi|\xi|^2) \text{ then } \mathcal{F}_0^{-1}f(x) = f(x). \quad (14.1) \quad \boxed{\text{E:Exp}}$$

Now, if we assume that $\Re z > 0$ and $A > 0$, then

$$A^{-z/2} = \frac{1}{\Gamma(z/2)} \int_0^\infty e^{-tA} t^{z/2} \frac{dt}{t}.$$

Using this we see that

$$(1 + |\xi|^2)^{-z/2} = \frac{1}{\Gamma(z/2)} \int_0^\infty e^{-t} e^{-t|\xi|^2} t^{z/2} \frac{dt}{t}. \quad (14.2) \quad \boxed{\text{E:BasselPreI}}$$

We now want to take the inverse Fourier transform of both sides in ξ . This reduces to just finding the inverse Fourier transform of $f(\xi) := \exp(-t|\xi|^2) = \exp\left(-\pi \left|\frac{\sqrt{t}\xi}{\sqrt{\pi}}\right|^2\right)$. Hence we can use (14.1) to see that

$$\begin{aligned} \mathcal{F}^{-1}f(x) &= \mathcal{F}_0^{-1}f\left(\frac{x}{2\pi}\right) \\ &= \frac{\exp\left(-\pi \left|\frac{x/(2\pi)}{\sqrt{t/\pi}}\right|^2\right)}{(t/\pi)^{d/2}} \\ &= \frac{\exp\left(-\pi \frac{|x|^2/4\pi^2}{t/\pi}\right)}{(t/\pi)^{d/2}} \\ &= \frac{\exp\left(-\frac{|x|^2}{4t}\right)}{(t/\pi)^{d/2}}. \end{aligned}$$

Combining this with (14.2) give us that

$$G_s(x) = \mathcal{F}^{-1}((1 + |\cdot|^2)^{-s/2}) = \frac{(\pi)^{-d/2}}{\Gamma(s/2)} \int_0^\infty e^{-t} e^{-|x|^2/(4t)} t^{(s-d)/2} \frac{dt}{t}.$$

Can you obtain the rate of $\mathcal{H}(t)$ as $t \rightarrow 0$? This is important.

15 Integrability conditions

We want to compare

1. $\Upsilon(0) < \infty$ and $\mathcal{H}(t) < \infty$
2. $\hat{f} \in L^p(\mathbb{R}^d)$ for $1 \leq p < d/(d-2)$.

(2) does not imply (1): Suppose that $\hat{f} \in L^p(\mathbb{R}^d)$ for $1 \leq p < d/(d-2)$. Then we may have as $|x| \rightarrow 0$ that $\hat{f}(x) \sim \frac{1}{|x|^{d-\delta}}$ where $0 < \delta < d$. As for the tails, the worst case scenario we may have is when $\hat{f} \sim \frac{1}{|x|^{d-\eta}}$ as $|x| \rightarrow \infty$ for some $0 < \eta < d$. So under these conditions at the origin and as $|x| \rightarrow \infty$, we need a $1 \leq p < d/(d-2)$ such that

$$p(d-\delta) < d, \quad \text{and} \quad p(d-\eta) > d,$$

which implies that we must have

$$\frac{d}{d-\eta} < p < \frac{d}{d-\delta} \quad \text{and} \quad 0 < \eta < \delta < 2.$$

However, this is not a problem and such an $\hat{f}$ exists. For example when $d = 3$, $\delta = 1.5$, $\eta = 1$, and

$$f(x) = \begin{cases} |x|^{\delta-d} = |x|^{-1.5} & |x| \leq 1 \\ |x|^{\eta-d} = |x|^{-2} & |x| \geq 10 \\ \text{continous and decreasing} & 1 < |x| < 10, \end{cases}$$

then if we choose a p such that

$$\frac{d}{d-\eta} = \frac{3}{2} < p < \frac{3}{1.5} = \frac{d}{d-\delta},$$

then $\hat{f} \in L^p(\mathbb{R}^d)$. For example, if $p = 1.75$ then $1.5 * 1.75 < 3$ and $2 * 1.75 > 3$ and so $\hat{f} \in L^p(\mathbb{R}^3)$. However this same $\hat{f}$ does not satisfy $\Upsilon(0) < \infty$ since around the origin, $\hat{f}(x)|x|^{-2}$ will not be integrable.

I have to read your arguments more carefully but it is interesting to know that the gap is huge here, I though the difference might be that we can allow logarithm correction to make $\Upsilon(0) < \infty$ but Tessitore's condition does not allow such arrangement.

16 Restarting the solution

16.1 Verification of restart

There is circular reasoning at (16.3). I will fix later

Suppose we have

$$u(t, x) = \begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta u(t, x) = b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot), \end{cases} \quad (16.1) \quad \boxed{\text{E: SHE}}$$

and

$$v(t, x) = \begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2}\Delta v(t, x) = b(x, v(t, x))\dot{W}_{t_0}(t, x) & x \in \mathbb{R}^d, t > 0, \\ v(0, \cdot) = u(t_0, \cdot), \end{cases} \quad (16.2) \quad \boxed{\text{E: SHErestart}}$$

where $\dot{W}_{t_0}(t, x) = \dot{W}(t + t_0, x)$. We want it to be the case that

$$v(t, x) = u(t + t_0, x). \quad (16.3) \quad \boxed{\text{E:tORestart}}$$

If we assume that (16.3) is true, then

$$\begin{aligned} v(t, x) &= \int_{\mathbb{R}^d} G(t, x - y)u(t_0, y)dy + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(y, v(s, y))W_{t_0}(ds, dy) \\ &= \int_{\mathbb{R}^d} G(t, x - y)u(t_0, y)dy + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(y, u(s + t_0, y))W_{t_0}(ds, dy) \end{aligned} \quad (16.4) \quad \boxed{\text{E:vsol}}$$

and because of (16.3), we would also like to have

$$v(t, x) = u(t + t_0, x) = \int_{\mathbb{R}^d} G(t + t_0, x - y)\mu(dy) + \int_0^{t+t_0} \int_{\mathbb{R}^d} G(t + t_0 - s, x - y)b(y, u(s, y))W(ds, dy). \quad (16.5) \quad \boxed{\text{E:vsolrestart}}$$

The first integral in (16.4) can be shown using Fubini's theorem to be written as

$$\int_{\mathbb{R}^d} G(t, x - y)u(t_0, y)dy = \int_{\mathbb{R}^d} G(t + t_0, x - y)\mu(dy) + \int_0^{t_0} \int_{\mathbb{R}^d} G(t + t_0 - s, x - y)b(y, u(s, y))W(ds, dy). \quad (16.6) \quad \boxed{\text{E:RestartHom}}$$

Thus to prove (16.5), it suffices to prove the following:

$$\begin{aligned} \int_0^{t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy) + \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y)b(y, u(s+t_0, y))W_t0(ds, dy) \\ = \int_0^{t+t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy). \end{aligned} \quad (16.7) \quad \boxed{\text{E:ChgOfVar}}$$

Intuitively, we would like the following to be true:

$$\int_0^t \int_{\mathbb{R}^d} G(t-s, x-y)b(y, u(s+t_0, y))W_t0(ds, dy) = \int_{t_0}^{t+t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy). \quad (16.8) \quad \boxed{\text{E:ChgOfVar2}}$$

Note that (16.8), would follow by the change of variable $s' = s + t_0$. Then plugging (16.8) into the LHS of (16.7) would give us

$$\text{LHS of (16.7)} = \int_0^{t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy) + \int_{t_0}^{t+t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy),$$

and lastly we would like for it to be the case that

$$\begin{aligned} \int_0^{t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy) + \int_{t_0}^{t+t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy) \\ = \int_0^{t+t_0} \int_{\mathbb{R}^d} G(t+t_0-s, x-y)b(y, u(s, y))W(ds, dy). \end{aligned} \quad (16.9) \quad \boxed{\text{E:SumOfInt}}$$

Then (16.8) and (16.9) would imply that (16.7) which in return would prove (16.5). Hence it suffices to prove (16.8) and (16.9). However, both

16.2 Assumption on the restarted initial condition

Also, in order to show the existence on an invariant measure, we need to assume that

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |\mu|)(x) < \infty \quad \text{for any } t \geq 0. \quad (16.10) \quad \boxed{\text{A:initialbdd}}$$

So if we want to show that our restarted solution, v , has an invariant measure, we will also need to show that

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |v(0, \cdot)|)(x) < \infty \quad \text{for any } t \geq 0.$$

which means that we need to show (almost surely) that

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |u(t_0, \cdot)|)(x) < \infty \quad \text{for any } t \geq 0. \quad (16.11) \quad \boxed{\text{A:RandomInit}}$$

Recall that

$$u(t_0, x) = \int_{\mathbb{R}^d} G(t_0, x-y)\mu(dy) + \int_0^{t_0} \int_{\mathbb{R}^d} G(t_0-s, x-y)b(y, u(s, y))W(ds, dy).$$

So $|u(t_0, \cdot)|$ is a processes and the absolute values add some difficulty. The comparison principle show that we may remove the absolute value signs for the δ_0 initial condition though. Thus we would just need to show

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * u(t_0, \cdot))(x) < \infty \quad \text{for any } t \geq 0.$$

We have that

$$\begin{aligned}
(G(t, \cdot) * u(t_0, \cdot))(x) &= \int_{\mathbb{R}^d} G(t, x - y) u(t_0, y) dy \\
&= \int_{\mathbb{R}^d} dy G(t, x - y) \left(\int_{\mathbb{R}^d} G(t_0, y - z) \delta_0(dz) + \int_0^{t_0} \int_{\mathbb{R}^d} G(t_0 - s, y - z) b(z, u(s, z)) W(ds, dz) \right) \\
&= \int_{\mathbb{R}^d} G(t, x - y) G(t_0, y) dy + \int_{\mathbb{R}^d} dy G(t, x - y) \int_0^{t_0} \int_{\mathbb{R}^d} W(ds, dz) G(t_0 - s, y - z) b(z, u(s, z)) \\
&= G(t + t_0, x) + \int_0^{t_0} \int_{\mathbb{R}^d} G(t + t_0 - s, x - z) b(z, u(s, z)) W(ds, dz),
\end{aligned}$$

where in the last line I assume that stochastic Fubini's theorem could be applied.

I answered this concern as I finished typing it. As long as we assume (16.10), then we will be able to show

$$\sup_{t \geq t_0} \mathbb{E} \left[\|u^\mu(t, \cdot)\|_\rho^2 \right] < \infty$$

which is all that is needed.

17 Fix for the random initial data

I previously proved that for $L_\rho^2(\mathbb{R}^d)$ initial conditions that

$$\mathbb{E} \int_0^1 \|Y^\varphi(s, \cdot)\|_\rho^q ds \leq K \|\varphi\|_\rho^q.$$

I think that for random initial conditions, $f(x)(\omega)$, that this can be extended to

$$\mathbb{E} \int_0^1 \|Y^f(s, \cdot)\|_\rho^q ds \leq K \mathbb{E} \left(\|f(\cdot)\|_\rho^q \right).$$

For random initial conditions we may still have the factorization formula

$$u(t, x) = (G(t, \cdot) * u^\mu(t_0, \cdot))(x) + \frac{\sin(\alpha\pi)}{\pi} F_\alpha^t(Y^{u^\mu(t_0, \cdot)})(x), \quad t \geq t_0.$$

We then define the set

$$\mathcal{K}(\Theta) := \left\{ (G(t_0, \cdot) * y)(x) + (F_\alpha^{t_0} h)(x), \quad \text{where } \mathbb{E}(\|y\|_\rho) \leq \Theta \text{ and } \mathbb{E}(\|h\|_{L^q(0,1,L_\rho^2)}) \leq \Theta \right\}.$$

We may then be able to continue the proof as in Tessitore Theorem 3.1. I need to still verify the details. This makes $\mathcal{K}$ a random set.

18 Lipschitz Constant

In our Invariant measure paper, we have only assumed that

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0. \quad (18.1) \quad \boxed{\text{E:LipCond}}$$

In your paper with Kunwoo Kim [CK19], you have that the Lipschitz constant must satisfy

$$\sup_{x \in \mathbb{R}^d} \frac{|b(x)|}{|x|} < \infty, \quad (18.2) \quad \boxed{\text{E:LipCondStr}}$$

which requires that $b(0) = 0$. We are also using Lemma 2.5 [CK19] which says that when $\Upsilon(0) < \infty$ and when $\nu > 2\gamma\Upsilon(0)$, then

$$H_\nu(t, \gamma) \leq \frac{\nu}{\nu - 2\gamma\Upsilon(0)}.$$

Recall that when considering the moment bound,

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(|\mu| * G(t, \cdot))(x)] H(t; \gamma_p)^{1/2},$$

that we have $\gamma_p = 32pL_b^2$. So we see that when $\nu = 2$ and when

$$L_b^2 \leq \frac{1}{32p\Upsilon(0)},$$

then $H(t; \gamma_p)$ will be bounded t . But this requires us to use the L_b from (18.2). If we just assume that $b(x)$ satisfies (18.1), then $b(0)$ may not be 0 and so (18.2) will not hold true and then we can't use Lemma 2.5 [CK19].

Should we then assume (18.2) in our invariant measure project?

This is a legal concern. It should not be a problem to extend our work with Kunwoo [CK19] to the case (18.1). You can either dig into our proofs to make this extension, or simply write the paper under (18.2). In the latter case, we can make a comment saying that we can handle the general case (18.1) as well.

References

- [BC16] [1] Balan, R. and Chen, Le Parabolic Anderson Model with Space-Time Homogeneous Gaussian Noise and Rough Initial Condition *J Theor Probab.* (2016)
- [CDfracheat15] [2] Chen, Le and Dalang, R. Moments, intermittency and growth indices for the nonlinear fractional stochastic heat equation *Stoch PDE: Anal Comp* (2015) 3:360397
- [CK19] [CK19] Chen, Le and Kunwoo Kim. Nonlinear stochastic heat equation driven by spatially colored noise: moments and intermittency. *Acta Math. Sci. Ser. B*, 39B (2019), No. 3, 645–668.
- [DZ92] [DZ92a] Da Prato, Giuseppe and Jerzy Zabczyk. *Stochastic equations in infinite dimensions*. Encyclopedia of Mathematics and its Applications, 44. Cambridge University Press, Cambridge, 1992. xviii+454 pp.
- [DZ96] [DZ96] Da Prato, Giuseppe and Jerzy Zabczyk. *Ergodicity for infinite-dimensional systems*. London Mathematical Society Lecture Note Series, 229. Cambridge University Press, Cambridge, 1996.
- [DZp92] [DZ92b] Da Prato, Giuseppe and Jerzy Zabczyk. *Invariant measures for semilinear stochastic equations*. Stochastic Anal. Appl. 10, 387-408.
- [DZ96] [DZ96] Da Prato, Giuseppe and Jerzy Zabczyk. *Ergodicity for infinite-dimensional systems*. London Mathematical Society Lecture Note Series, 229. Cambridge University Press, Cambridge, 1996.
- [Gra14] [Gra14] L. Grafakos (2000) Modern Fourier Analysis { *Graduate Texts in Mathematics*}
- [TZ98] [TZ98] Tessitore, Gianmario and Jerzy Zabczyk. Invariant measures for stochastic heat equations. *Probab. Math. Statist.* 18 (1998), no. 2, Acta Univ. Wratislav. No. 2111, 271–287.
- [Winklebauer] [3] Winklebauer. Moments and Absolute Moments of the Normal Distribution. *Arxiv* <https://arxiv.org/abs/1209.4340>.